

- 14** The decomposition of a gas, A, undergoes a 3-step mechanism. The rate constants of the three steps are given by k_1 , k_2 , and k_3 respectively.

The overall rate equation for the reaction is given by

$$\text{rate} = \frac{k_1 k_3 [A]^2}{k_3 + k_2 [A]}$$

Consider the two statements about the order of reaction for the decomposition of A at constant temperature.

- 1 At very low pressures of A, the reaction is second order with respect to A.
- 2 At very high pressures of A, the reaction is first order with respect to A.

Which statements are correct?

- A** Both statements are correct.
- B** Neither statement is correct.
- C** Statement 1 only is correct.
- D** Statement 2 only is correct.